Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems

Complete project status — what is done, what failed, what happens next

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The project in one sentence.

Stating a privacy guarantee over PEOPLE instead of over RECORDS costs accuracy — and that cost is not a fixed number. It grows with how many records each person contributes, then levels off. This work measures that curve, proves the measurement with a control built into the same experiment, and is now testing it on a second, independent clinical corpus against a forecast frozen in advance.

1. At a glance

Category	Count	Note
Experiments completed	24	All finished; none abandoned mid-run
Succeeded outright	14	Including every engineering objective
Characterised negative results	5	Reported rather than dropped
Mixed or inconclusive	3	WEAR accuracy, WEAR fusion, attribution
Superseded	1	End-to-end budget sweep, kept as comparison arm
Cost priced, not passed/failed	1	Rigorous public pretraining

The four objectives from the January 2026 proposal

ID	Objective	Target	VitalDB	WEAR
O1	Uplink traffic reduction	≥ 50%	**90.82% PASS**	**96.08% PASS**
O1b	Round latency reduction	> 0%	**16.93% PASS**	**21.68% PASS**
O2a	Privacy budget (train + conformal)	ε ≤ 1	**0.9705 PASS**	**0.9656 PASS**
O2b	Accuracy loss under DP	≤ 2 pp	**−0.92 pp PASS**	**1.967 pp INCONCL.**
O3a	Inference latency P95	< 100 ms	**4.58 ms PASS**	**1.22 ms PASS**
O3b	Successful updates	> 99%	**100% PASS**	**99.94% PASS**
O4	Energy saving per round	≥ 20%	**72.30% PASS**	**76.84% PASS**

VitalDB 7/7 PASS. WEAR 6 PASS + 1 INCONCLUSIVE. O2a is a COMPOSED budget — training 0.7705 plus conformal release 0.2000 — stricter than the common practice of reporting training alone.

The honest reading of that table.

Every engineering objective was met, most by a wide margin. They turned out to be the easy part. Nothing in the January proposal anticipated the finding that now carries the paper — the k-scaling law, the patient-vs-record privacy unit, or the membership-inference audit. The project outgrew its own proposal.

2. What was done, and what it measured

Phase 4 — Simulation infrastructure

E1 · NS-3 packet-level network simulation

NS-3 v3.46.1 · 9 IoT devices, 2 edge servers, 1 cloud · device→edge 54 Mbps, edge→cloud 1 Gbps at 5 ms · LogDistance propagation, 5 Mbps background traffic

Quantity	VitalDB	WEAR
Round latency, flat → hierarchical	90.5 → 75.2 ms	104.5 → 81.8 ms
Latency reduction	16.93%	21.68%
Per-update reliability (2 hops)	100%	99.94%
Round uplink / downlink time	33.97 s / 26.85 s	—

Result: the edge tier reduces both traffic and round latency under real packet-level measurement, not an analytical estimate.

E2 · CloudSim Plus compute and energy simulation

CloudSim Plus v8.5.6 · IoT 300 MIPS / 512 MB / 1–3 W · edge 8000 MIPS / 16 GB / 50–200 W · GPU overhead modelled on edge servers only

Scenario	Energy per round	Uplink
CentralizedFL	1266.25 J	73.85 MB
HFL_Proposed (VitalDB)	**−72.3%**	see E10
HFL_Proposed (WEAR)	**−76.8%**	see E10

Result: 72.3% and 76.8% energy saving against a 20% target. A second figure of 93.5% exists from a bytes→Joules proxy but is arithmetically identical to communication reduction, so it is not an independent energy measurement and is not quoted.

Phase 5 — Model, privacy and deployment

E3 · Encoder pretraining: proxy vs rigorous

Mode	Effect on AUROC	Privacy validity
Proxy / oracle	baseline	**ε NOT valid end-to-end**
Rigorous disjoint public split	−0.022	**valid**

Result: a valid end-to-end guarantee costs 0.022 AUROC. That gap is the number to report, not the proxy alone.

E4 · Baseline suite B0–B5

Baseline	Test accuracy	Test macro-AUC	Note
B0 Centralized	0.5787	0.8211	Upper reference (pooled data)
B1 Local-Only	0.4532	0.7511	Lower reference, no collaboration
B2 FedAvg	0.6543	0.8795	Best non-private baseline
B3 FedProx	0.6570	0.8747	No advantage over FedAvg here
B4 DP-FedAvg (ε≈1)	**0.2159**	**0.6443**	COLLAPSES. The like-for-like DP reference
B5 MOON	0.3934	0.7418	Underperforms on this task

Result: B4 is the critical row. Applying DP the conventional way at the same budget destroys the model — which is what makes E5 meaningful.

E5 · Head-only DP fine-tuning — the ε=1 collapse fix

20 rounds, 5 edge rounds · δ=1e-5, C=1.0 · head = 560,645 / 2,157,766 params (26.0%) · 3 seeds

Recipe	σ	Test AUC	Round time
End-to-end DP-SGD	11.3281	**0.562**	1046 s
Head-only DP fine-tuning	11.3281	**0.8742 ± 0.0072**	50 s

The noise was never the problem.

σ is identical in both rows. What changed is optimisation geometry: a near-linear probe over frozen pretrained features tolerates σ≈11.3, while training from scratch does not. Also ~20× faster per round. This reframes the standard "DP destroys utility" complaint.

E7 · ε sweep, rigorous head recipe

ε	σ	AUROC	AUPRC
0.5	5.92	0.8287	0.5110
1	3.24	0.8319	0.5464
2	1.88	0.8323	0.5381
4	1.21	0.8356	0.5590
8	0.85	0.8342	0.5544
∞ (non-private)	—	0.8158	0.5204

Result: utility is flat from ε=0.5 to ε=8, and every private arm beats the non-private one. DP acts as a regulariser here. The utility floor lies below the budget the system ships at.

E8 · Objective 2b — multi-seed accuracy cost

Dataset	Cost (pp)	95% CI	Verdict
VitalDB (n=5)	−0.92	[−2.82, +0.99]	**PASS**
WEAR (n=15)	1.967	[1.760, 2.173]	**INCONCLUSIVE**

Result: VitalDB shows no measurable cost. WEAR's interval straddles the 2 pp threshold, so it can be neither passed nor failed. The seed count was pre-registered at 15 and deliberately not extended — extending it after seeing the result would be p-hacking.

E10 · Gradient compression and uplink measurement

Measurement	Bytes per round
Full-model FP32 FedAvg	258,931,920
Head-only, uncompressed	67,277,400
Observed, post-fix	**16,821,300**

Defect found and fixed: the compressor took a fixed k = numel × sparsity of EVERY tensor, so frozen encoders — whose delta is exactly zero — contributed 20% of their elements as explicit zeros. Omitting all-zero tensors alone accounted for a 78.87% reduction. Final: 90.82% (VitalDB), 96.08% (WEAR) against a 50% target.

E11 · ONNX export and inference latency

1000 timed runs after 100 warmup, batch 1 · path = preprocessing → ONNX forward → conformal set → serialise

Precision	Provider	Model	Inference P95	Conformal	Total P95	QoS

FP32	CPU	10.64 MB	13.259	0.079	16.363 ms	**PASS**
FP32	GPU (CUDA)	10.64 MB	20.088	0.096	22.476 ms	**PASS**
INT8	CPU	6.55 MB	27.877	0.090	31.028 ms	**PASS**
INT8	GPU (CUDA)	6.55 MB	70.020	0.112	73.150 ms	**PASS**

Headline objective figures — 4.58 ms P95 (VitalDB), 1.22 ms (WEAR) — come from the final objective run on the current corpora.

E13 · Patient-level vs record-level DP

Both arms at matched budget · 5 seeds · VitalDB: 48,712 windows / 6,257 cases, mean 7.79 windows per case, 100% of cases contribute more than one window

Result: the upgrade is priced at 1.77× σ. Because every patient contributes more than one record, record-level and patient-level are DIFFERENT STATEMENTS for every individual in the dataset. This is the premise the whole project rests on, and it is measured, not assumed.

E20 · Feature attribution

Recipe	Mask share of attribution
full_dp	28.2%
patient-DP	22.0%
sample-DP	21.6%
full_nodp	19.9%
head_nodp	15.8%

Result: DP shifts the model's attention toward MISSINGNESS — under noise it leans more on which measurements exist than on their values, monotonically with how much DP is applied. Rank correlation ρ=0.95 private vs not. Evidence, not proof.

E22 · Component ablation A0–A6

Arm	Configuration	AUROC	AUPRC	ε
A0	Full system (reference)	0.8541 ± 0.0035	0.5960	0.7705
A1	No C1 — flat tier (no edge)	0.8484 ± 0.0057	0.5911	0.7704
A2	No C2 — no DP	0.8307 ± 0.0060	0.5136	∞

Important: A1–A4 never produced a result until four silent defects were fixed on 2026-08-16 — a dead --local_epochs argument, a uint8-hardcoded quantiser, a summary reading test_* from a CSV holding only val_*, and a 1e-4 verdict threshold.

E23 · NS-3 stress tests

Result: the system holds across round counts up to 250. This sweep also exposed a DEFINITION DEFECT: update reliability had been computed as hop_success ^ n_rounds — the probability that ALL rounds succeed — which decays with round count, so any system fails a ">99%" target for a long enough run. Per-update reliability is 99.8001%; the whole-run figure is now emitted under the distinct name run_reliability_all_rounds_pct.

3. The headline result

E18 · The k-scaling law

All 24 configurations: k ∈ {1,2,4,8} × 3 seeds × 2 arms · per-step budget and units-per-step held constant

k	Record arm AUPRC	Patient arm AUPRC	Gap	95% CI	p
1	0.5015	0.5067	−0.0052	[−0.0664, +0.0561]	0.75
2	0.5590	0.5084	**+0.0507**	[+0.0417, +0.0596]	0.0017
4	0.5924	0.5092	**+0.0832**	[+0.0694, +0.0971]	0.0015
8	0.5955	0.5089	**+0.0866**	[+0.0603, +0.1129]	0.0049

Why this result is strong — two reasons.

1. IT CARRIES ITS OWN CONTROL. At k=1 a patient IS a record, so the two guarantees are mathematically identical and the arms must collapse. They do (p = 0.75). A validated null sitting inside the same experiment as the positive effects. That is what licenses reading the divergence at larger k as a property of the privacy unit rather than an artefact of the apparatus.

2. THE MECHANISM IS ASYMMETRIC. The patient arm is flat in k (0.4967–0.5176 at every k); the record arm rises (0.5015 → 0.5955). The widening gap is THE RECORD ARM IMPROVING, not the patient arm degrading. Patient-level noise scales with per-patient contribution, so extra records buy the patient arm nothing.

Honest limits. AUROC shows the same monotone shape (+0.0006, +0.0094, +0.0133, +0.0132) but no k reaches significance — every interval includes zero. AUPRC is the resolving metric; AUROC is directionally consistent but underpowered at n=3. Accuracy is pinned at 0.8005 in every patient row — an uncalibrated 0.5 threshold on imbalanced data — and MUST NOT BE QUOTED for these arms.

4. What failed — reported in full

The credibility of the positive results depends on these being on the record.

N-A · The refuted central hypothesis (E14, E16, E24)

The project's original central prediction was that record-level DP would leave patients exposed (attack AUC ≈ 0.70) while patient-level DP defended them (≈ 0.50). A gap of ~0.20 was predicted.

Arm	Best attack AUC (95% CI)	Difficulty floor	Margin
Positive control — head-only, no DP	0.6277 [0.5905, 0.6649]	0.5626	**+0.0627**
Record-level DP	0.5477 [0.4876, 0.6079]	0.5252	+0.0226
Patient-level DP	0.5224 [0.5175, 0.5274]	0.5167	+0.0057

Retested at ε=16 (E24), where noise is 13× lower (σ 11.33 → 0.85): record-level 0.5465 against a floor of 0.5743 — BELOW ITS OWN FLOOR, i.e. confound rather than leakage.

Verdict: REFUTED, twice, each time with a passing positive control.

WHY IT FAILED. Not a weak attack — the control cleared a gate set BEFORE running. DP-SGD at ε=1 is simply very strong: σ=11.33 leaves almost no membership signal in EITHER arm. A supporting clue confirms it: on the patient arm, LiRA scored WORSE than a plain loss threshold (−0.0071). That is what a defence with nothing left to leak looks like.

HOW THIS MUST ALWAYS BE WRITTEN: the large predicted effect is refuted; a small effect is NOT excluded — the interval is ±0.060 wide at n=5. Never write "there is no effect".

E15 · The unexpected finding inside the failure

Recipe	Attack AUC	Member / non-member loss
full-model, no DP	0.5122	1.81 / 2.31
head-only, no DP	**0.5847**	0.013 / 4.95
full-model, DP	0.5206	6.93 / 7.46

LEAKAGE IS A PROPERTY OF THE RECIPE, NOT THE TASK. Head-only fine-tuning — our own efficiency mechanism — memorises members far harder than full-model training (p = 1e−9, sd 0.0018 over 6 runs). A genuine caveat to the parameter-efficient DP fine-tuning literature, and a caveat against our own recipe. It became claim N-A′ when N-A failed.

Other characterised negatives

Item	Outcome
E9 SADP — adaptive per-group budget	**Underperforms uniform DP at equal ε BY CONSTRUCTION. The cross-group composition tax drives σ from 11.3 to 70. Lesson: splitting a budget across groups is paid for at a punitive rate under composition.**
E12 INT8 edge quantisation	**1.90× slower on CPU, 3.25× slower on GPU than FP32, and misses the 4 MB target at 6.55 MB. Two causes: dynamic quantisation only accelerates MatMul/Gemm while this model is dominated by convolutions and a recurrent path; and on CUDA the quantised ops have no GPU kernel, so the runtime inserts 3180 Memcpy nodes — exactly the 3.25× penalty. Deploy FP32.**
ASWA aggregation	**+0.0005 AUROC. Deleted from the codebase.**

E19 Fusion gain, corrected	**At 2 seeds: +0.0411 AUROC, +0.1062 AUPRC. At 5 seeds: +0.0370 and +0.0671 — AUPRC FELL 37%. Only the 5-seed figures may be quoted.**
Legacy corpus (ECG + chest X-ray)	**Withdrawn entirely. Its modalities were paired by a construction rule (CXR bucket = f(ECG label)) rather than by provenance, so no result on it was evidence of fusion.**
WEAR accuracy objective (O2b)	**Inconclusive. Seed count pre-registered at 15 and deliberately not extended.**
E19 WEAR fusion	**At ceiling. Best single modality already 0.9976, so the task has no headroom. Structurally unable to corroborate the fusion claim.**

E19 · Multimodal fusion — the mixed result

Arm	VitalDB AUROC	WEAR AUROC
Modality A only (signals / inertial)	0.8283 ± 0.0062	0.9976
Modality B only (clinical / video)	0.7680 ± 0.0033	0.9957
Fused	**0.8653 ± 0.0052**	0.9980
Gain over best unimodal	**+0.0370 (5.9× seed sd)**	**+0.0004 (ceiling)**

5. The outstanding gap — and what is being done about it

The one question the project cannot currently answer.

Every privacy claim rests on ONE DATASET FROM ONE HOSPITAL. VitalDB is a single institution and its nine "clients" are a partition we created ourselves. WEAR cannot help — 22 people, one client each, no patient contributing enough records for the k-law to even be defined.

A reviewer will ask, and it is the right question:
"Is this a property of patient-level private federated learning, or a property of your dataset?"

The answer: eICU-CRD

Property of eICU-CRD	Why the research needs it
208 real hospitals	Clients become actual institutions rather than a partition we invented
~139,000 patients	Different demographics, care patterns, equipment. If the law survives that, it is structural
Many records per patient	VitalDB tops out near k=8. eICU stretches the curve into a range VitalDB cannot reach. THIS IS THE DECISIVE PROPERTY.
Same modality structure	Time-series vitals + static clinical. No new encoder needed
Credentialed + CITI trained	Reviewers trust PhysioNet; the access process is itself evidence of proper conduct

Access granted. BRSET also granted, parked until eICU is complete.

The rule that makes this experiment worth running

Freeze the prediction before touching the data.

We will NOT simply run the same experiment and add another table. That is worth little.

Instead: fit the k-curve on VitalDB, write down a NUMERIC prediction of the gap at eICU's k, FREEZE THAT PREDICTION BEFORE TOUCHING THE DATA, then test it.

Why frozen first: if we fit the curve, look at eICU, and then write the prediction, no reviewer can distinguish that from retrofitting — and neither can we.

The frozen forecast (committed before the build)

Fitted on VitalDB data alone:

$$\text{gap}(k) = 0.0878 \cdot (1 - \exp(-0.883 \cdot (k-1)))$$

A = 0.0878 ± 0.0035 · b = 0.883 ± 0.129 · RMSE = 0.0028

k	Predicted gap	Acceptance band (±0.030)
1	+0.000	−0.030 … +0.030 (control)
2	+0.052	+0.022 … +0.082
4	+0.082	+0.052 … +0.112
8	+0.088	+0.058 … +0.118
16	+0.088	+0.058 … +0.118
32	+0.088	+0.058 … +0.118

The curve SATURATES — 90% of the ceiling by k = 3.6. That is the risky content of the forecast, and it is testable ONLY at k > 8, which is precisely what VitalDB cannot do and eICU can.

6. What happens next

Execution order — every step has a gate

Step	Experiment	Gate
E0	**Commit the pre-registration**	**Must precede the build. Non-negotiable.**
E1	Verify schema against real headers	Fix any mismatched constant first
E2	Build corpus	—
E3	Validate corpus	Do not proceed past a CAPACITY failure
E4	Ceiling check (single-modality arms)	**Stop if better arm > 0.95 AUROC — the check WEAR failed**
E5	Pretrain encoders, disjoint population	So eICU does not inherit the oracle caveat
E6	Main operating point	ε = 0.80 + 0.20 conformal, 10 rounds, τe = 5
E7	Objective verification O1–O4	—
E8	**k-scaling — THE experiment**	**k ∈ {1,2,4,8,16,32}. k=1 is the built-in control; k=16, 32 are the point of the corpus**
E9	LiRA + non-private positive control	A null without a passing control is inadmissible
E10	Component ablation A0–A5	—
E11	Uplink / compression	Re-measure, do not assume

How each remaining gap closes

Gap	Closed by
Every claim rests on one single-institution dataset	E8 on 208 real hospitals
Is the k-law structural or dataset-specific? Unanswerable	E8 against the frozen numeric forecast
Curve cannot be stretched past VitalDB's k	E8 at k = 16, 32
Encoders pretrained in oracle mode; ε=1 not end-to-end	E5 with the disjoint population
Fusion single-domain; gain shrank at 5 seeds	E4 at 5 seeds, reported as a setting result
MIA null needs statistical power	E9 at ~139,000 patients
Federation is constructed, not natural	E10 — hospitals taken as given

Independent of eICU — still open

Item	Why it matters
Window-count-normalised patient clipping	**THE ONLY REMAINING CANDIDATE FOR AN ORIGINAL MECHANISM rather than a measurement. A patient's gradient is summed over k windows then clipped to a fixed bound, systematically down-weighting long recordings. Normalising before clipping should correct it. Needs no new data.**
Conformal coverage validation	The guarantee is implemented and its cost composed into the reported ε, but empirical coverage has never been measured
Worst-case MIA reporting (TPR at 0.1% FPR)	Average AUC hides the tail. The confident identifications are the real clinical disclosure. Machinery already exists; cost is low
Mask-channel ablation	Would convert the E20 attribution evidence into proof
Multi-seed baselines	Would put confidence intervals on E4

7. How the research ends

The eICU experiment has two possible outcomes, and BOTH ARE PUBLISHABLE. That is by design — it is what makes the experiment worth running rather than decorative.

Outcome	What it means
The prediction lands	The cost of a patient-level guarantee is shown to be STRUCTURAL rather than dataset-specific, and the work delivers a transferable design rule: given your k, this is what patient-level ε=1 will cost you. That is what a practitioner actually needs, and it has not been published.
The prediction misses	The work has found a BOUNDARY CONDITION, with a dated forecast proving the story was not adjusted after the fact. Also publishable, and visibly honest.

Either way, the paper is stronger.

A paper reporting that two independent clinical datasets were tested — one reproducing the result and the other exposing a limitation — is MORE publishable than one in which everything conveniently succeeded.

What is genuinely ours, and what is not

Prior art — do not claim	Genuinely this work's contribution
Patient-level / user-level DP (McMahan et al., 2018)	**THE k-SCALING LAW ITSELF — the measured relationship between records-per-person and the cost of a person-level guarantee. Not previously published.**
Head-only / parameter-efficient DP fine-tuning (Li, Yu et al., 2021)	**THE SELF-CONTROLLING EXPERIMENT DESIGN — the k=1 null built into the same experiment as the effect.**
Exponential-mechanism DP quantile release (Smith, 2011)	**AN AUDIT THAT CAN RETURN AN INFORMATIVE NEGATIVE — every attack reported alongside a positive control on a configuration known to memorise.**
Hierarchical federated learning (HED-FL and others)	**A FINDING THAT ARGUES AGAINST OUR OWN RECIPE — head-only training memorises harder (0.5847 vs 0.5122), reported rather than buried.**

The contributions are MEASUREMENTS AND AUDITING PRACTICE in a setting where they have not been measured — not new mechanisms. The manuscript must not say "we propose patient-level DP".

The final claim

Stating your privacy guarantee over people instead of over records costs accuracy — and that cost is not a constant. It grows with how many records each person contributes, then saturates. Here is the curve, here is the control that proves the curve is real, and here is the independent corpus that says it is not just our data.

Sources: Research Dossier Parts 1–3; Complete Experiment Log E1–E24; document/01_planning/plan.md rev 3; document/01_planning/PREREGISTRATION_EICU.md; Phase Reports 1–6. All values quoted are from measured artifacts on disk.